Mohammad Sadra Moazzen

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OBJECTIVE

Aspiring Computational Neuroscientist and NeuroAI Researcher with expertise in cell and molecular biology, data science, and artificial intelligence. Passionate about applying advanced analytical techniques and machine learning to unravel complex biological questions, particularly in computational neuroscience and neuroimaging.

EDUCATION

Master of Science in Cell and Molecular Biology 2025 - 2027
Hakim Sabzevari University

MicroMaster in Artificial Intelligence 2023 - 2026
Sharif University of Technology GPA: 3.5 / 4.0

MicroMaster in Data Science 2023 - 2024
Sharif University of Technology GPA: 3.66 / 4.0

Bachelor of Science in Cell and Molecular Biology 2021 - 2025
Ferdowsi University of Mashhad GPA: 3.64 / 4.0

SKILLS

Programming Languages	Python, MATLAB, R, C/C++
Data Science Tools	SQL, NumPy, Pandas, Matplotlib, Seaborn, Bokeh
Machine Learning Frameworks	TensorFlow, PyTorch, Scikit-learn, OpenCV
Neuroimaging Tools	SPM, CONN, FREESURFER, FSL, Nilearn
Computational Neuroscience Tools	NWB, NeuroDSP, MNE, NetworkX, Elephant

INTERESTS

Computational Neuroscience, NeuroAI, Machine Learning

RESEARCH EXPERIENCE

MIT (CompNeuro) August 2024 - February 2025
Researcher (Remote)

- Working with Dr. Nima Dehghani on one research paper:
- *Decoding Mouse Sleep Stage Neural Dynamics with Time Delay Stability, Phase Synchronization, Neural Granger Causality, and Auxiliary-Driven Self-Supervised Embedding*

Research Institute of Biotechnology July 2023 - August 2026
Undergraduate Researcher (In-Person)

- Worked with Dr. Fatemeh B. Rassouli on two research papers:
- *Maximizing Efficacy: Machine Learning-Driven Dose-Time Predictions for Coumarin Therapies.*
- *In Vitro Insights: Machine Learning-Driven Evaluation of Esculetin's Anticancer Action.*

Interdisciplinary Schools September 2023 - November 2025
Researcher (Remote)

- Worked with Dr. Abolhasan Rezaeyan on one research paper:
- *Temporal Lobe Epilepsy Connectivity: Insights from MRI and Machine Learning.*

PUBLICATIONS

PUBLICATIONS

Machine learning-guided dose-time optimization and experimental validation enhance coumarin therapeutics in oncology
Scientific Reports — *Published*

- Used machine learning to identify optimal dose-time combinations for enhancing the anticancer efficacy of natural coumarins.

Pooled Evidence Supports Esculetin as a Promising Natural Coumarin for Cancer Therapy
Journal of Pharmaceutical Innovation — *Published*

- Conducted a systematic evaluation of in vitro evidence on the anticancer effects of esculetin across human cancers.

Temporal Lobe Epilepsy Connectivity: Insights from MRI and Machine Learning Under Review

- Applied machine learning to classify patients with temporal lobe epilepsy using structural and functional connectivity features derived from MRI and resting-state fMRI.

PROJECTS

Enhancing Deep Learning Robustness in Microscopy Images Project — July 2024
Programmer

- Improved U-Net's robustness in microscopy image segmentation by addressing noise and artifacts.
- Tools: Python, PyTorch, OpenCV, Matplotlib.
- Project repository: GitHub

Sternberg Memory Task — March 2024
Programmer

- Simulated a classic psychological experiment on short-term memory using Python.
- Tools: Python, Pygame, Matplotlib.
- Project repository: GitHub

TEACHING EXPERIENCE

Neuroscience and AI — November 2025
Shahid Beheshti University of Medical Sciences

- Delivered six in-depth sessions on the intersection of neuroscience and artificial intelligence, highlighting core neural mechanisms, recent research advances, and their applications in cognitive modeling and AI systems.

Machine Learning in Neuroimaging — February 2025 - June 2025
Interdisciplinary Schools

- Designed and taught a course on ML applications in neuroimaging.

Machine Learning Techniques — December 2024 - February 2025
Interdisciplinary Schools

- Taught a course on Machine Learning techniques and their applications.

Data Science — July 2024 - October 2024
Interdisciplinary Schools

- Taught a Data Science course covering NumPy, Pandas, Matplotlib, and Seaborn.

TEACHING ASSISTANT POSITIONS

Computational Neuroscience August 2026
NeuroCONNECT

- Led a group of 15 students over 3 weeks in hands-on computational neuroscience education. Guided interactive online classrooms and mentored research projects. Topics covered included spiking neuron models, synaptic plasticity, neural dynamics, reinforcement learning, probabilistic models, and deep learning in neuroscience. Provided expert support in computational modeling and neuroscience methodologies, fostering a collaborative and inclusive learning environment.

Computational Neuroscience July 2025
The Neuromatch Academy

- Led a group of 14 students over 3 weeks in hands-on computational neuroscience education. Guided interactive online classrooms and mentored research projects. Topics covered included spiking neuron models, synaptic plasticity, neural dynamics, reinforcement learning, probabilistic models, and deep learning in neuroscience. Provided expert support in computational modeling and neuroscience methodologies, fostering a collaborative and inclusive learning environment.

Python for Non-Engineers December 2024 - February 2025
Interdisciplinary Schools

- Supported the course by introducing Python fundamentals, facilitating hands-on exercises, guiding discussions, and ensuring accessibility for participants from diverse academic backgrounds.

VOLUNTEER POSITIONS

Course Coordinator and Content Lead: Python in Neuroimaging (fMRI) July–October 2025
Interdisciplinary Schools

- Oversaw lesson design and delivery quality for an fMRI-focused Python module by coordinating with the instructor and assistant through regular meetings, ensuring pedagogical coherence, technical accuracy, and alignment with best practices in neuroimaging education.

Collaborator: Software and Programming Languages October 2024 - July 2026
Interdisciplinary Schools

- Contributed to software development and optimization projects, utilizing expertise in programming languages and frameworks to deliver scalable, high-quality solutions.

Course Developer & Instructor: Computational Neuroscience September 2024 - July 2026
Interdisciplinary Schools

- Developed and instructed a course in computational neuroscience, focusing on the design of comprehensive educational materials.

Mentor: Neuroimaging Group July 2024 - August 2026
Interdisciplinary Schools

- Mentored a four-person neuroimaging team, providing guidance to enhance learning, support research initiatives, and foster innovative problem solving.

Editor-in-Chief & Writer December 2022 - December 2023
Ferdowsi University of Mashhad

- Volunteered as Editor-in-Chief of the Biology Scientific Association, leading content creation and editorial processes.

CERTIFICATIONS

Deep Learning July 2024
Neuromatch Academy

Machine Learning Specialization April 2024
DeepLearning.AI

Understanding the Brain: The Neurobiology of Everyday Life August 2023
University of Chicago

CS50's Introduction to Programming with Python August 2023
CS50

Data Analyst Professional Certificate October 2023
IBM

CS50's Introduction to Databases with SQL November 2023
CS50

Cognitive Neuroscience November 2023
Shahid Beheshti University

Computational Neuroscience July 2023
Neuromatch Academy